RETAIL SALES PERFORMANCE ANALYTICS

End-to-End Business Intelligence Project

Presenters:

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BUSINESS PROBLEM

4 key questions we aim to answer:

1) Why do certain **product categories** underperform?

2) How do return rates impact **profit margins**?

3) What are the purchasing behaviors of different **demographics**?

4) How do **seasonal trends** affect shipping and sales?

PROJECT OVERVIEW

MAIN GOAL:

Deliver a governed, interactive Power BI foundation for data-driven retail decisions.

Scope of Analysis

Analyzing historical retail transactions to unlock business value.

- **Sales & Profitability:** Tracking revenue drivers and profit margins.
- **Regional Performance:** Monitoring geographic trends and shipping efficiency.
- **Customer Segmentation:** Understanding demographic purchasing behaviors.

PROJECT OBJECTIVES

Revenue Tracking:	Analyze sales drivers using Time Intelligence (YoY & MoM Growth).
Product Profitability:	Identify Top N Products driving the highest profit margins.
Operational Efficiency:	Measure and optimize return rates and shipping logistics.
Data Governance:	Implement Row-Level Security (RLS) for secure regional access.

TECHNOLOGY STACK

An integrated pipeline from raw data extraction to interactive business intelligence.

Excel & Power Query:
Data Extraction, Cleansing, and Folder Consolidation.

Power BI & Security:
Interactive Dashboards, UX Design, and Row-Level Security (RLS).

Data Modeling & DAX:
Building Star Schema, Relationships, and calculating Advanced KPIs.

IMPLEMENTATION PROCESS

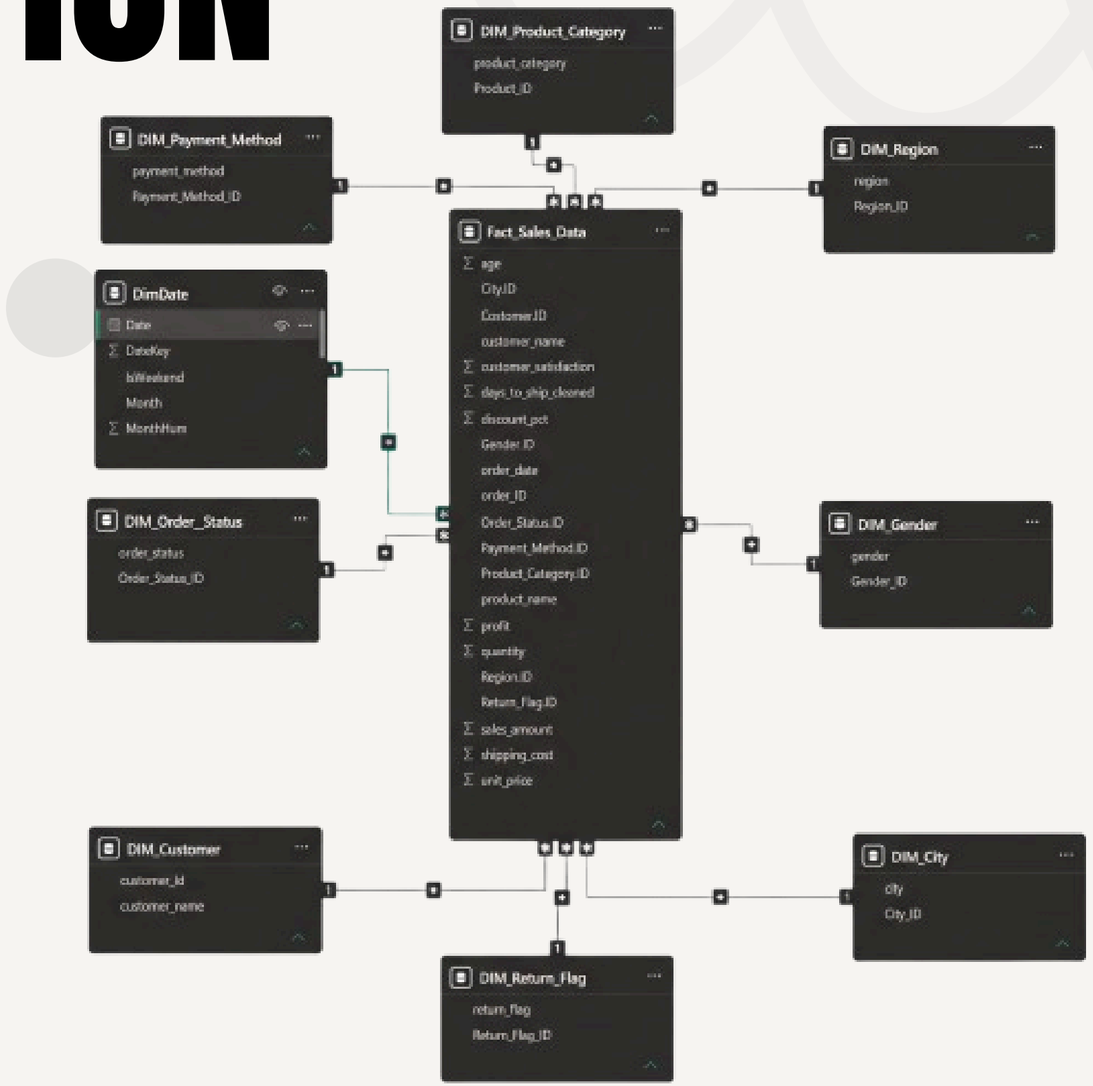
Transforming raw data into a structured, high-performance analytical model.

01 Data Cleaning (Power Query)

02 Data Consolidation

03 Data Modeling (Star Schema)

04 Advanced DAX Engine



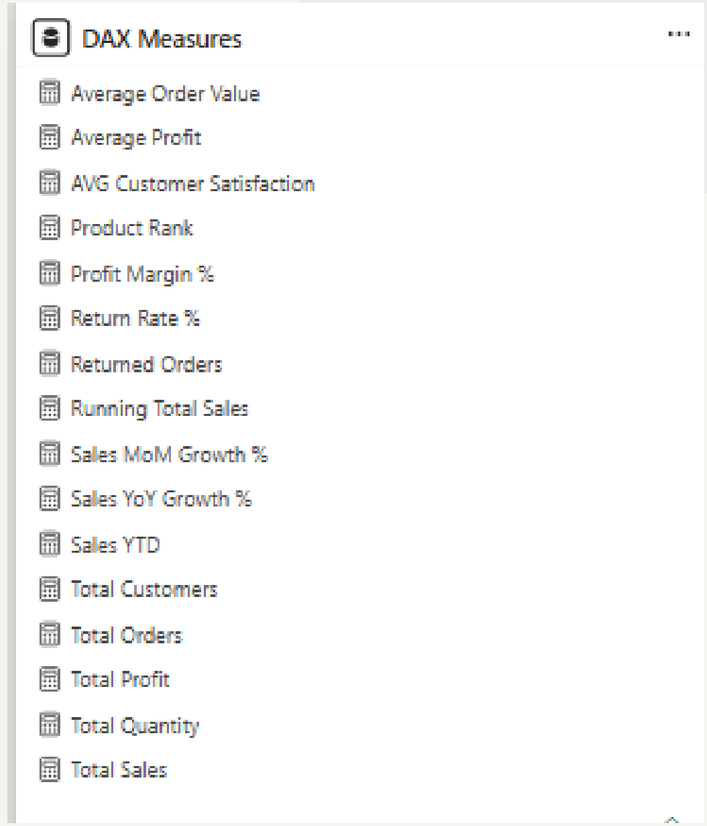
ADVANCED DAX ENGINE

Powering the dashboard with dynamic business logic and complex calculations.

Time Intelligence:
Calculated Sales YTD, YoY Growth %, and MoM trends to track historical performance.

Performance Metrics:
Developed precise measures for Profit Margin %, Average Order Value (AOV), and Return Rates.

Advanced Ranking:
Implemented dynamic Top N ranking to instantly identify best-selling products and top regions.



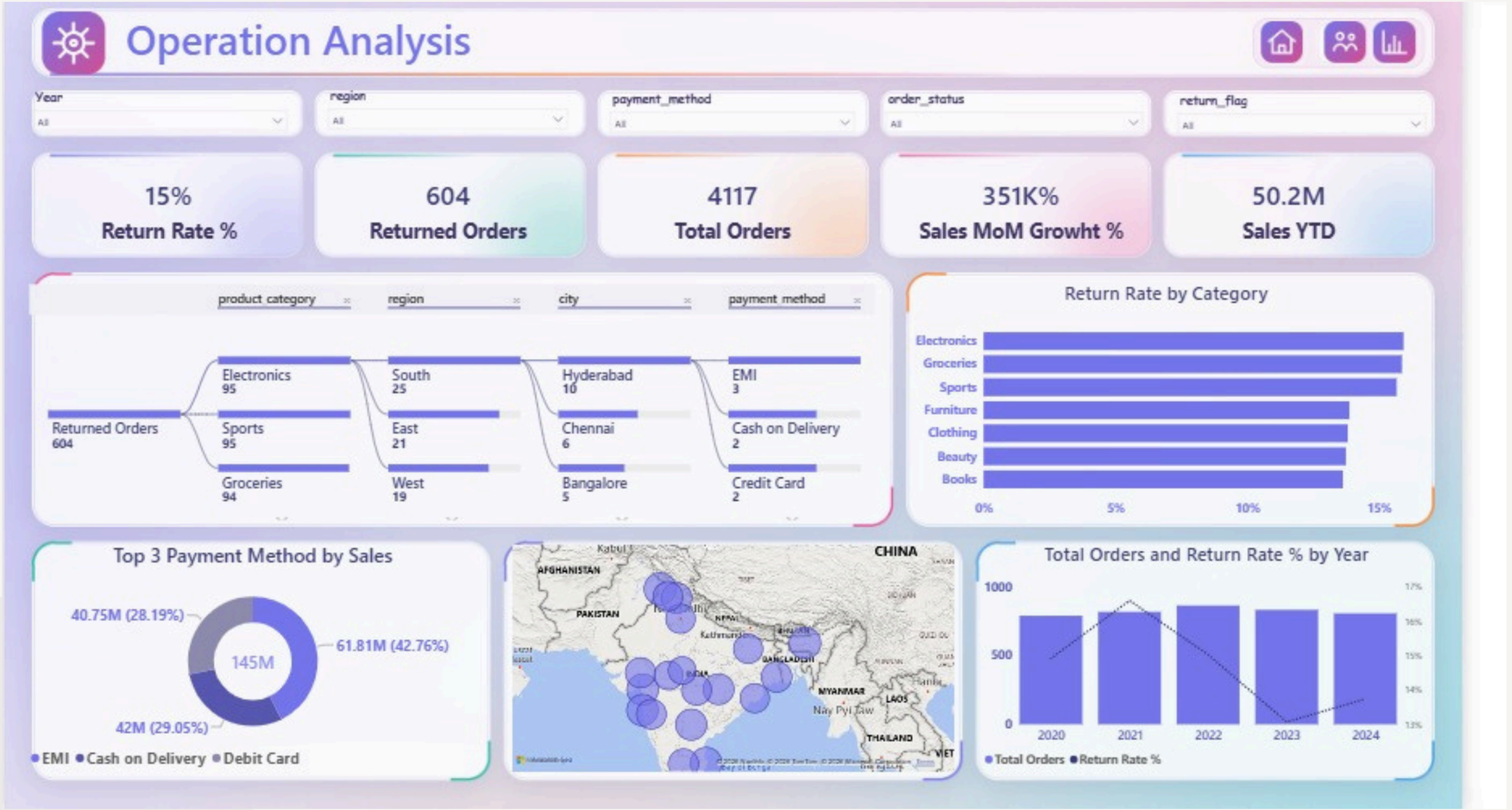
SALES PERFORMANCE

Kochi, Amritsar, and Chennai lead city performance, while a concentrated set of product categories drives the sales mix.



OPERATION ANALYSIS

Payment mix, category return rates, and order trends reveal where operational friction is concentrated.



CUSTOMER INSIGHTS

Regional distribution, gender mix, and category preferences create a practical base for customer segmentation.



REGION ANALYSIS

Compare customer counts and return rates by region and city to locate expansion potential and service issues.



city	product_category	Total Orders	Returned Orders	Return Rate %
Bhubaneswar	Beauty	2	1	50.00%
Bhubaneswar	Books	1	1	100.00%
Bhubaneswar	Clothing	1		
Bhubaneswar	Electronics	4		
Bhubaneswar	Furniture	1		
Guwahati	Beauty	2		
Guwahati	Clothing	1		
Guwahati	Electronics	1		
Guwahati	Furniture	1		

BUSINESS INSIGHTS & ACTION PLAN

1. Revenue Growth: Achieved $259.7M total sales with a strong 24.0% YoY growth, driven primarily by the West and South regions.

2. Operational Bottleneck: Overall return rate sits at 15%, with the "Electronics" category driving the highest volume of returned orders.

Recommendation: Investigate Electronics quality control and shipping damages

3. Customer Value: Average Order Value (AOV) is exceptionally high at $63.08K.

Recommendation: Implement loyalty programs for top-tier clients (e.g., Kavya Kapoor) to sustain high profit margins.

KEY LEARNING

1. Data Foundation is Everything: A robust data model (Star Schema) and clean data are critical for accurate DAX calculations and high-performance reporting.

2. Design Drives Adoption: Applying UI/UX principles ensures that complex data becomes easy to understand and navigate for business users.

3. Governance is Non-Negotiable: Implementing Row-Level Security (RLS) is essential to protect sensitive data and restrict access based on user roles.



THANK YOU

Opening the floor for questions and discussion

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